

# **DATABASE DESIGN DOCUMENT**

**Group no. - 3**

**Project Title**

**LabStock AI – Intelligent Laboratory Inventory Management System**

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# 1. Introduction

LabStock AI is an intelligent inventory management system designed specifically for **clinical diagnostic laboratories**. The system is developed to manage laboratory supplies, monitor stock levels, manage suppliers and purchase orders, maintain inventory transaction history, and support intelligent demand forecasting and reorder recommendations.

The database acts as the central storage component of the system. It maintains information related to laboratory supplies, suppliers, users, inventory transactions, purchase orders, forecasts, and reorder recommendations.

The database is designed using a relational database approach to maintain data consistency, reduce redundancy, and establish clear relationships between different entities.

## 2. Purpose of the Database

The main purpose of the database is to provide a structured and reliable way to store and manage all inventory-related information of the laboratory.

The database will:

- Store laboratory supply information.
- Maintain current stock quantities.
- Define reorder points and safety stock.
- Maintain supplier information.
- Manage purchase orders.
- Track ordered and received quantities.
- Maintain historical inventory transactions.
- Provide historical data for demand forecasting.
- Store predicted demand and stockout risk.
- Store intelligent reorder recommendations.
- Support role-based access to the system.
- Provide data required for dashboards and reports.

### 3. Laboratory Scope

For this project, we are considering a **Clinical Diagnostic Laboratory**.

The laboratory may handle different types of materials such as:

- Diagnostic reagents
- CBC reagent kits
- Glucose test kits
- Blood collection tubes
- Urine sample containers
- Test tubes
- Pipette tips
- Microscope slides
- Disposable gloves
- PPE
- Other laboratory consumables

The scope of LabStock AI is limited to **inventory and procurement management**.

The following are outside the scope of this project:

- Patient registration
- Patient medical records
- Doctor management
- Appointment management
- Billing
- Diagnosis
- Detailed diagnostic test-result management

### 4. Database Objectives

The major objectives of the database are:

1. To maintain accurate laboratory inventory records.
2. To monitor the quantity of laboratory supplies.
3. To identify low-stock items using reorder points.
4. To maintain supplier information.
5. To manage purchase orders and their status.
6. To track partially fulfilled purchase orders.
7. To maintain historical inventory movement.
8. To provide historical data for demand forecasting.
9. To store forecasting results.
10. To generate and store intelligent reorder recommendations.

11. To maintain data integrity through keys and constraints.
12. To provide secure and role-based access to information.

## 5. Technology Used

The proposed database system uses the following technologies:

| Component            | Technology                     |
|----------------------|--------------------------------|
| Database             | Microsoft SQL Server           |
| Backend              | ASP.NET Core                   |
| Programming Language | C#                             |
| ORM                  | Entity Framework Core          |
| Authentication       | ASP.NET Core Identity          |
| Database Design      | Relational Database / ER Model |

## 6. Database Entities

The main entities identified for LabStock AI are:

1. **Users**
2. **Suppliers**
3. **LabSupplies**
4. **InventoryTransactions**
5. **PurchaseOrders**
6. **PurchaseOrderItems**
7. **DemandForecasts**
8. **ReorderRecommendations**

These entities are designed based on the functional requirements of the project.

# 7. Table Design

## 7.1 Users

The Users entity stores information about people who access the LabStock AI system. Different users can have different roles and permissions.

| Column      | Data Type     | Key | Description                         |
|-------------|---------------|-----|-------------------------------------|
| UserId      | NVARCHAR(450) | PK  | Unique user ID                      |
| FullName    | NVARCHAR(100) |     | Name of the user                    |
| Email       | NVARCHAR(256) |     | User email                          |
| Role        | NVARCHAR(50)  |     | User role                           |
| IsActive    | BIT           |     | Indicates whether account is active |
| CreatedDate | DATETIME2     |     | Account creation date               |

Authentication-related information such as password hashes is handled by ASP.NET Core Identity.

## 7.2 Suppliers

The Suppliers table stores information about companies or individuals supplying laboratory materials.

| Column        | Data Type     | Key | Description             |
|---------------|---------------|-----|-------------------------|
| SupplierId    | INT           | PK  | Unique supplier ID      |
| SupplierName  | NVARCHAR(150) |     | Name of supplier        |
| ContactPerson | NVARCHAR(100) |     | Supplier contact person |
| ContactEmail  | NVARCHAR(256) |     | Email address           |
| Phone         | NVARCHAR(20)  |     | Contact number          |
| Address       | NVARCHAR(300) |     | Supplier address        |
| IsActive      | BIT           |     | Supplier status         |
| CreatedDate   | DATETIME2     |     | Record creation date    |

## 7.3 LabSupplies

The LabSupplies table is the main inventory table. It stores information about each laboratory supply and its current stock.

| Column         | Data Type     | Key | Description                  |
|----------------|---------------|-----|------------------------------|
| SupplyId       | INT           | PK  | Unique supply ID             |
| SupplyName     | NVARCHAR(150) |     | Name of laboratory supply    |
| Category       | NVARCHAR(50)  |     | Type/category of supply      |
| Description    | NVARCHAR(500) |     | Description of supply        |
| Unit           | NVARCHAR(30)  |     | Box, kit, bottle, piece etc. |
| QuantityOnHand | INT           |     | Current available stock      |
| ReorderPoint   | INT           |     | Minimum stock threshold      |

|             |               |    |                          |
|-------------|---------------|----|--------------------------|
| SafetyStock | INT           |    | Required buffer quantity |
| ImageURL    | NVARCHAR(500) |    | Image of supply          |
| SupplierId  | INT           | FK | Associated supplier      |
| IsActive    | BIT           |    | Supply status            |
| CreatedDate | DATETIME2     |    | Creation date            |
| UpdatedDate | DATETIME2     |    | Last update              |

For example, if a glucose reagent has a quantity of 12 and its reorder point is 15, the system will identify it as a low-stock item.

## 7.4 InventoryTransactions

The InventoryTransactions table maintains the historical movement of laboratory supplies.

It records activities such as consumption, restocking and stock adjustments.

| Column          | Data Type     | Key | Description                           |
|-----------------|---------------|-----|---------------------------------------|
| TransactionId   | BIGINT        | PK  | Unique transaction ID                 |
| SupplyId        | INT           | FK  | Related laboratory supply             |
| TransactionType | NVARCHAR(30)  |     | Consumption, Restocking or Adjustment |
| Quantity        | INT           |     | Quantity involved                     |
| TransactionDate | DATETIME2     |     | Date and time of transaction          |
| UserId          | NVARCHAR(450) | FK  | User performing transaction           |
| ReferenceId     | INT           |     | Related order/reference if applicable |
| Remarks         | NVARCHAR(500) |     | Additional information                |

This table is especially important for the AI component because historical consumption data can be used for demand forecasting.

## 7.5 PurchaseOrders

The PurchaseOrders table stores the main information about purchase orders created for laboratory supplies.

| Column               | Data Type     | Key | Description                                          |
|----------------------|---------------|-----|------------------------------------------------------|
| PurchaseOrderId      | INT           | PK  | Unique purchase order ID                             |
| SupplierId           | INT           | FK  | Supplier associated with order                       |
| CreatedBy            | NVARCHAR(450) | FK  | User who created order                               |
| OrderDate            | DATETIME2     |     | Date of order                                        |
| ExpectedDeliveryDate | DATETIME2     |     | Expected delivery date                               |
| Status               | NVARCHAR(30)  |     | Pending, Partially Fulfilled, Fulfilled or Cancelled |
| Notes                | NVARCHAR(500) |     | Additional order information                         |
| CreatedDate          | DATETIME2     |     | Record creation date                                 |

## 7.6 PurchaseOrderItems

The PurchaseOrderItems table stores the individual supplies included in a purchase order.

Separating the order and its items allows one purchase order to contain multiple laboratory supplies.

| Column              | Data Type | Key | Description            |
|---------------------|-----------|-----|------------------------|
| PurchaseOrderItemId | INT       | PK  | Unique order-item ID   |
| PurchaseOrderId     | INT       | FK  | Related purchase order |
| SupplyId            | INT       | FK  | Supply being ordered   |
| QuantityOrdered     | INT       |     | Quantity ordered       |



|                  |               |                            |
|------------------|---------------|----------------------------|
| QuantityReceived | INT           | Quantity actually received |
| UnitPrice        | DECIMAL(10,2) | Price per unit             |

For example, if 100 blood collection tubes are ordered but only 60 are received, the system stores:

**QuantityOrdered = 100**

**QuantityReceived = 60**

Therefore, the order can be identified as **Partially Fulfilled**.

## 7.7 DemandForecasts

The DemandForecasts table stores the output of the demand forecasting component.

| Column            | Data Type     | Key | Description                 |
|-------------------|---------------|-----|-----------------------------|
| ForecastId        | BIGINT        | PK  | Unique forecast ID          |
| SupplyId          | INT           | FK  | Supply being forecast       |
| ForecastStartDate | DATE          |     | Start of forecast period    |
| ForecastEndDate   | DATE          |     | End of forecast period      |
| PredictedQuantity | DECIMAL(12,2) |     | Predicted demand            |
| StockoutRisk      | NVARCHAR(20)  |     | Low, Medium or High         |
| ModelName         | NVARCHAR(100) |     | Model used for forecasting  |
| GeneratedDate     | DATETIME2     |     | Date forecast was generated |

Historical inventory consumption data can be used to generate the predicted quantity.

## 7.8 ReorderRecommendations

The ReorderRecommendations table stores intelligent recommendations generated by the system.

| Column                  | Data Type      | Key | Description                                  |
|-------------------------|----------------|-----|----------------------------------------------|
| RecommendationId        | BIGINT         | PK  | Unique recommendation ID                     |
| SupplyId                | INT            | FK  | Related laboratory supply                    |
| ForecastId              | BIGINT         | FK  | Forecast used                                |
| CurrentQuantity         | INT            |     | Current stock                                |
| PredictedDemand         | DECIMAL(12,2)  |     | Predicted requirement                        |
| SafetyStock             | INT            |     | Safety stock                                 |
| PendingPurchaseQuantity | INT            |     | Quantity already ordered                     |
| RecommendedQuantity     | INT            |     | Suggested reorder quantity                   |
| RiskLevel               | NVARCHAR(20)   |     | Low, Medium or High                          |
| RecommendationReason    | NVARCHAR(1000) |     | Reason for recommendation                    |
| Status                  | NVARCHAR(30)   |     | Pending, Accepted, Rejected or ConvertedToPO |
| CreatedDate             | DATETIME2      |     | Recommendation date                          |

This table allows the system to store not only the recommended quantity but also the reason and risk level associated with the recommendation.

## 8. Primary Keys and Foreign Keys

Primary keys uniquely identify records within a table.

Foreign keys establish relationships between tables.

| Table                  | Primary Key         | Foreign Keys              |
|------------------------|---------------------|---------------------------|
| Users                  | UserId              | —                         |
| Suppliers              | SupplierId          | —                         |
| LabSupplies            | SupplyId            | SupplierId                |
| InventoryTransactions  | TransactionId       | SupplyId, UserId          |
| PurchaseOrders         | PurchaseOrderId     | SupplierId, CreatedBy     |
| PurchaseOrderItems     | PurchaseOrderItemId | PurchaseOrderId, SupplyId |
| DemandForecasts        | ForecastId          | SupplyId                  |
| ReorderRecommendations | RecommendationId    | SupplyId, ForecastId      |

## 9. Entity Relationships

The major relationships in the database are:

### **Supplier → LabSupplies**

One supplier can provide multiple laboratory supplies.

**Relationship: 1 : Many**

### **LabSupply → InventoryTransactions**

One laboratory supply can have many inventory transactions.

**Relationship: 1 : Many**

**User → InventoryTransactions**

One user can perform multiple inventory transactions.

**Relationship: 1 : Many**

**Supplier → PurchaseOrders**

One supplier can have multiple purchase orders.

**Relationship: 1 : Many**

**PurchaseOrder → PurchaseOrderItems**

One purchase order can contain multiple items.

**Relationship: 1 : Many**

**LabSupply → PurchaseOrderItems**

One laboratory supply can appear in multiple purchase-order items.

**Relationship: 1 : Many**

**LabSupply → DemandForecasts**

One laboratory supply can have multiple forecasts for different periods.

**Relationship: 1 : Many**

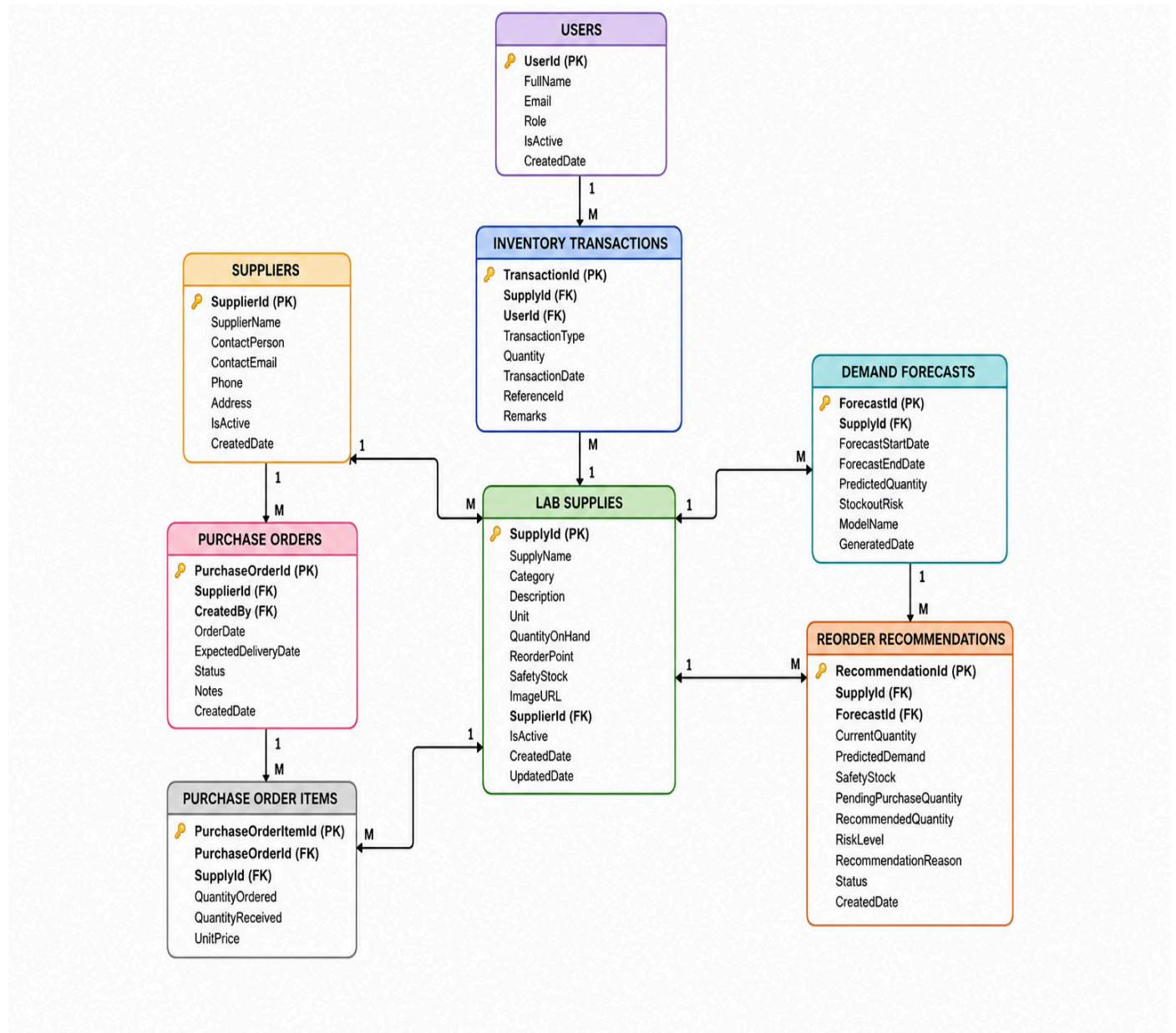
**DemandForecast → ReorderRecommendations**

A forecast can be associated with reorder recommendations.

**Relationship: 1 : Many**

# 10. ER Diagram

The overall database relationship can be represented as:



# 11. Database Constraints

The following constraints will be applied to maintain data integrity:

## **Primary Key Constraint**

Every table will have a primary key that uniquely identifies each record.

## **Foreign Key Constraint**

Foreign keys will ensure that relationships between entities remain valid.

## **NOT NULL Constraint**

Important fields such as supply name, supplier name and quantities should not be left empty.

## **UNIQUE Constraint**

Fields such as user email can be required to be unique.

## **CHECK Constraint**

Quantity-related fields should not contain invalid negative values.

For example:

QuantityOnHand  $\geq 0$

QuantityOrdered  $> 0$

QuantityReceived  $\geq 0$

# 12. Normalization

The database design follows normalization principles to reduce data duplication and maintain consistency.

## **First Normal Form — 1NF**

Each field contains a single atomic value and repeating groups are avoided.

## **Second Normal Form — 2NF**

Non-key attributes depend on the complete primary key.

### Third Normal Form — 3NF

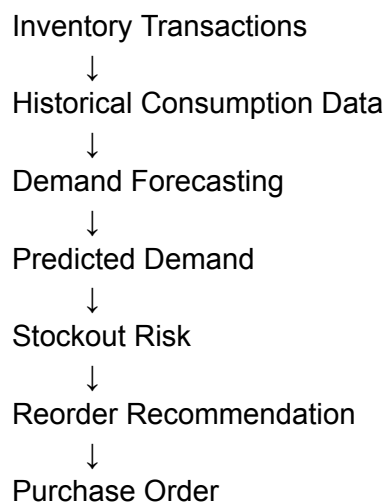
Non-key attributes depend only on the primary key and unnecessary transitive dependencies are avoided.

The separation of `PurchaseOrders` and `PurchaseOrderItems` is an example of designing the database to avoid unnecessary duplication.

## 13. AI/ML Data Flow

The database supports the intelligent features of LabStock AI through historical inventory data.

The process is:



The `InventoryTransactions` table provides historical consumption data. This information can be used by the forecasting component to predict future demand.

The forecast results are stored in `DemandForecasts`.

Based on current stock, predicted demand, safety stock and pending orders, the system generates a recommendation which is stored in `ReorderRecommendations`.

## 14. Sample Data

The database can contain laboratory supplies such as:

| Supply Name            | Category   | Quantity | Reorder Point |
|------------------------|------------|----------|---------------|
| CBC Reagent Kit        | Reagent    | 25       | 10            |
| Glucose Test Kit       | Test Kit   | 12       | 15            |
| Blood Collection Tube  | Consumable | 80       | 30            |
| Urine Sample Container | Consumable | 100      | 40            |
| Pipette Tips           | Consumable | 35       | 50            |

For example, the Glucose Test Kit has 12 units available while its reorder point is 15. Therefore, the system can identify it as a low-stock item.

## 15. Role-Based Access

The system will provide different levels of access depending on the user's role.

| Role              | Main Responsibilities                 |
|-------------------|---------------------------------------|
| Administrator     | Manage users and system settings      |
| Inventory Manager | Manage supplies and monitor stock     |
| Procurement Staff | Manage suppliers and purchase orders  |
| Laboratory Staff  | Record consumption and view inventory |

Role-based access helps prevent unauthorized users from performing restricted operations.



## 16. Requirement-to-Database Mapping

| Functional Requirement    | Database Component                  |
|---------------------------|-------------------------------------|
| User Management           | Users                               |
| Inventory Management      | LabSupplies                         |
| Stock Monitoring          | LabSupplies + InventoryTransactions |
| Supplier Management       | Suppliers                           |
| Purchase Order Management | PurchaseOrders + PurchaseOrderItems |
| Partial Order Fulfillment | PurchaseOrderItems                  |
| Inventory History         | InventoryTransactions               |
| Demand Forecasting        | DemandForecasts                     |
| Stockout Risk             | DemandForecasts                     |
| Intelligent Reordering    | ReorderRecommendations              |
| Reporting                 | Multiple database tables            |

## 17. Security and Backup

The database will use authentication and authorization mechanisms to ensure that only authorized users can access restricted functionality.

Role-based access will be used to control operations performed by administrators, inventory managers, procurement staff and laboratory staff.

Regular database backups should be maintained to prevent loss of important inventory and procurement information.

## 18. Conclusion

The proposed database design provides the relational foundation for the LabStock AI system. It manages laboratory supplies, suppliers, inventory transactions, purchase orders and user information while also supporting the intelligent components of the system.

The addition of historical inventory transactions allows the system to use consumption data for demand forecasting. Forecast results can then be used to identify stockout risks and generate intelligent reorder recommendations.

Therefore, the database supports both the **traditional inventory-management requirements** and the **AI-based features** of LabStock AI for a clinical diagnostic laboratory.